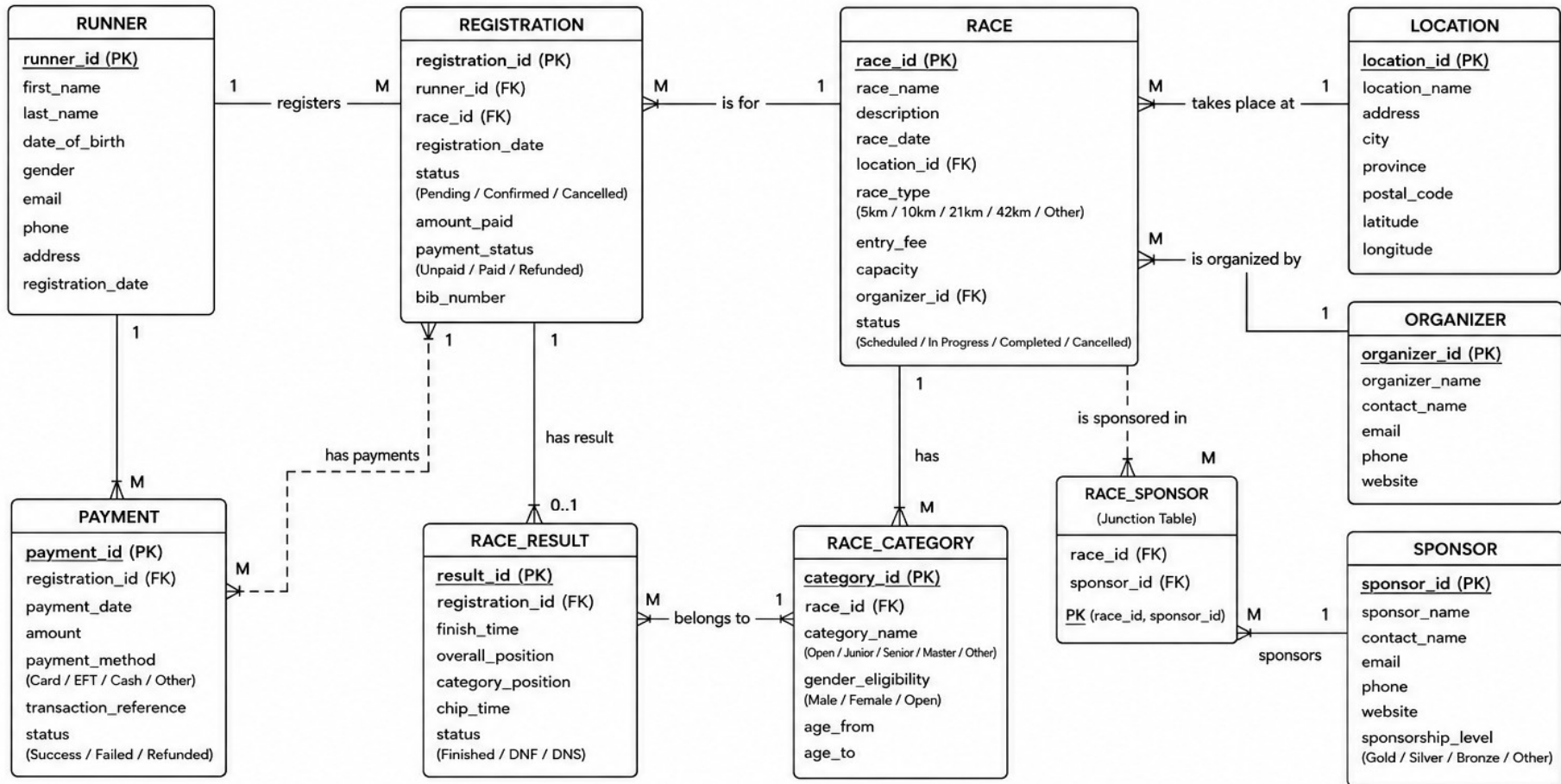


RaceDay – Entity Relationship Diagram (ERD)



LEGEND

PK	Primary Key
FK	Foreign Key
1	One
M	Many
0..1	Zero or One
—	One-to-Many
---	Many-to-Many

NOTES

- RUNNER 1—M REGISTRATION:** A runner can register for many races. Each registration belongs to one runner.
- REGISTRATION M—1 RACE:** A race can have many registrations. Each registration is for one race.
- REGISTRATION 1—M PAYMENT:** A registration can have one or more payments.
- REGISTRATION 0..1—1 RACE_RESULT:** A registration can have zero or one result.
- RACE_RESULT M—1 RACE_CATEGORY:** A result belongs to one category.
- RACE 1—M RACE_CATEGORY:** A race can have many categories.
- RACE M—1 LOCATION:** Many races can be held at one location.
- RACE M—1 ORGANIZER:** Many races can be organized by one organizer.
- RACE M—M SPONSOR:** Many races can have many sponsors and many sponsors can sponsor many races (via RACE_SPONSOR).

RACEDAY – ERD

Model Version:	1.0
Date:	18 May 2025
Designed By:	RaceDay Team
Description:	Core data model for the RaceDay event management system.

Section C

/*

RACE_DAY DATABASE

SQL Server / SSMS

Section C - SQL Database Script

*/

IF DB_ID('Race_Day') IS NULL

BEGIN

CREATE DATABASE Race_Day;

END

GO

USE Race_Day;

GO

-- DROP TABLES

IF OBJECT_ID('dbo.RACE_RESULT', 'U') IS NOT NULL

DROP TABLE dbo.RACE_RESULT;

IF OBJECT_ID('dbo.PAYMENT', 'U') IS NOT NULL

```
DROP TABLE dbo.PAYMENT;
```

```
IF OBJECT_ID('dbo.REGISTRATION', 'U') IS NOT NULL
```

```
DROP TABLE dbo.REGISTRATION;
```

```
IF OBJECT_ID('dbo.RACE_SPONSOR', 'U') IS NOT NULL
```

```
DROP TABLE dbo.RACE_SPONSOR;
```

```
IF OBJECT_ID('dbo.RACE_CATEGORY', 'U') IS NOT NULL
```

```
DROP TABLE dbo.RACE_CATEGORY;
```

```
IF OBJECT_ID('dbo.RACE', 'U') IS NOT NULL
```

```
DROP TABLE dbo.RACE;
```

```
IF OBJECT_ID('dbo.SPONSOR', 'U') IS NOT NULL
```

```
DROP TABLE dbo.SPONSOR;
```

```
IF OBJECT_ID('dbo.LOCATION', 'U') IS NOT NULL
```

```
DROP TABLE dbo.LOCATION;
```

```
IF OBJECT_ID('dbo.ORGANIZER', 'U') IS NOT NULL
```

```
DROP TABLE dbo.ORGANIZER;
```

```
IF OBJECT_ID('dbo.RUNNER', 'U') IS NOT NULL
```

```
    DROP TABLE dbo.RUNNER;
```

```
GO
```

```
-- 1. RUNNER
```

```
CREATE TABLE RUNNER
```

```
(
```

```
    runner_id INT IDENTITY(1,1) NOT NULL,
```

```
    first_name NVARCHAR(50) NOT NULL,
```

```
    last_name NVARCHAR(50) NOT NULL,
```

```
    date_of_birth DATE NOT NULL,
```

```
    gender NVARCHAR(10) NOT NULL,
```

```
    email NVARCHAR(100) NOT NULL,
```

```
    phone NVARCHAR(20) NOT NULL,
```

```
    address NVARCHAR(200) NOT NULL,
```

```
    registration_date DATE NOT NULL
```

```
    CONSTRAINT DF_RUNNER_registration_date
```

```
    DEFAULT CONVERT(DATE, GETDATE()),
```

```
    CONSTRAINT PK_RUNNER
```

PRIMARY KEY (runner_id),

CONSTRAINT UQ_RUNNER_email

UNIQUE (email),

CONSTRAINT CK_RUNNER_gender

CHECK (gender IN ('Male', 'Female', 'Other'))

);

GO

-- 2. ORGANIZER

CREATE TABLE ORGANIZER

(

organizer_id INT IDENTITY(1,1) NOT NULL,

organizer_name NVARCHAR(100) NOT NULL,

contact_name NVARCHAR(100) NOT NULL,

email NVARCHAR(100) NOT NULL,

phone NVARCHAR(20) NOT NULL,

website NVARCHAR(200) NULL,

CONSTRAINT PK_ORGANIZER

PRIMARY KEY (organizer_id),

CONSTRAINT UQ_ORGANIZER_email

UNIQUE (email)

);

GO

-- 3. LOCATION

CREATE TABLE LOCATION

(

location_id INT IDENTITY(1,1) NOT NULL,

location_name NVARCHAR(100) NOT NULL,

address NVARCHAR(200) NOT NULL,

city NVARCHAR(80) NOT NULL,

province NVARCHAR(80) NOT NULL,

postal_code NVARCHAR(10) NOT NULL,

latitude DECIMAL(9,6) NULL,

longitude DECIMAL(9,6) NULL,

CONSTRAINT PK_LOCATION

PRIMARY KEY (location_id),

```
CONSTRAINT UQ_LOCATION_name_address
```

```
    UNIQUE (location_name, address),
```

```
CONSTRAINT CK_LOCATION_latitude
```

```
    CHECK (latitude IS NULL OR latitude BETWEEN -90 AND 90),
```

```
CONSTRAINT CK_LOCATION_longitude
```

```
    CHECK (longitude IS NULL OR longitude BETWEEN -180 AND 180)
```

```
);
```

```
GO
```

```
-- 4. RACE
```

```
CREATE TABLE RACE
```

```
(
```

```
    race_id INT IDENTITY(1,1) NOT NULL,
```

```
    race_name NVARCHAR(150) NOT NULL,
```

```
    description NVARCHAR(500) NOT NULL,
```

```
    race_date DATE NOT NULL,
```

```
    location_id INT NOT NULL,
```

```
    race_type NVARCHAR(10) NOT NULL,
```

entry_fee DECIMAL(10,2) NOT NULL,

capacity INT NOT NULL,

organizer_id INT NOT NULL,

status NVARCHAR(20) NOT NULL

CONSTRAINT DF_RACE_status

DEFAULT 'Scheduled',

CONSTRAINT PK_RACE

PRIMARY KEY (race_id),

CONSTRAINT UQ_RACE_name_date

UNIQUE (race_name, race_date),

CONSTRAINT CK_RACE_type

CHECK (race_type IN ('5km', '10km', '21km', '42km', 'Other')),

CONSTRAINT CK_RACE_entry_fee

CHECK (entry_fee >= 0),

CONSTRAINT CK_RACE_capacity

CHECK (capacity > 0),

CONSTRAINT CK_RACE_status

CHECK

(

status IN

(

'Scheduled',

'In Progress',

'Completed',

'Cancelled'

)

),

CONSTRAINT FK_RACE_LOCATION

FOREIGN KEY (location_id)

REFERENCES LOCATION(location_id),

CONSTRAINT FK_RACE_ORGANIZER

FOREIGN KEY (organizer_id)

REFERENCES ORGANIZER(organizer_id)

);

GO

-- 5. RACE_CATEGORY

CREATE TABLE RACE_CATEGORY

(

category_id INT IDENTITY(1,1) NOT NULL,

race_id INT NOT NULL,

category_name NVARCHAR(50) NOT NULL,

gender_eligibility NVARCHAR(10) NOT NULL,

age_from INT NOT NULL,

age_to INT NOT NULL,

CONSTRAINT PK_RACE_CATEGORY

PRIMARY KEY (category_id),

CONSTRAINT UQ_RACE_CATEGORY_name

UNIQUE (race_id, category_name),

CONSTRAINT CK_RACE_CATEGORY_gender

CHECK

(

gender_eligibility IN

('Male', 'Female', 'Open')

),

CONSTRAINT CK_RACE_CATEGORY_age_from

CHECK (age_from >= 0),

CONSTRAINT CK_RACE_CATEGORY_age_to

CHECK (age_to >= age_from),

CONSTRAINT FK_RACE_CATEGORY_RACE

FOREIGN KEY (race_id)

REFERENCES RACE(race_id)

);

GO

-- 6. REGISTRATION

CREATE TABLE REGISTRATION

(

registration_id INT IDENTITY(1,1) NOT NULL,

runner_id INT NOT NULL,

race_id INT NOT NULL,

registration_date DATE NOT NULL

CONSTRAINT DF_REGISTRATION_registration_date

DEFAULT CONVERT(DATE, GETDATE()),

status NVARCHAR(20) NOT NULL

CONSTRAINT DF_REGISTRATION_status

DEFAULT 'Pending',

amount_paid DECIMAL(10,2) NOT NULL

CONSTRAINT DF_REGISTRATION_amount_paid

DEFAULT 0.00,

payment_status NVARCHAR(20) NOT NULL

CONSTRAINT DF_REGISTRATION_payment_status

DEFAULT 'Unpaid',

bib_number INT NULL,

CONSTRAINT PK_REGISTRATION

PRIMARY KEY (registration_id),

CONSTRAINT UQ_REGISTRATION_runner_race

UNIQUE (runner_id, race_id),

CONSTRAINT UQ_REGISTRATION_bib

UNIQUE (race_id, bib_number),

CONSTRAINT CK_REGISTRATION_status

CHECK

(

status IN

('Pending', 'Confirmed', 'Cancelled')

),

CONSTRAINT CK_REGISTRATION_payment_status

CHECK

(

payment_status IN

('Unpaid', 'Paid', 'Refunded')

),

CONSTRAINT CK_REGISTRATION_amount_paid

CHECK (amount_paid >= 0),

CONSTRAINT CK_REGISTRATION_bib

CHECK (bib_number IS NULL OR bib_number > 0),

CONSTRAINT FK_REGISTRATION_RUNNER

FOREIGN KEY (runner_id)

REFERENCES RUNNER(runner_id),

CONSTRAINT FK_REGISTRATION_RACE

FOREIGN KEY (race_id)

REFERENCES RACE(race_id)

);

GO

-- 7. PAYMENT

CREATE TABLE PAYMENT

(

payment_id INT IDENTITY(1,1) NOT NULL,

registration_id INT NOT NULL,

payment_date DATETIME2 NOT NULL

CONSTRAINT DF_PAYMENT_payment_date

DEFAULT SYSDATETIME(),

amount DECIMAL(10,2) NOT NULL,
payment_method NVARCHAR(20) NOT NULL,
transaction_reference NVARCHAR(100) NOT NULL,

status NVARCHAR(20) NOT NULL

CONSTRAINT DF_PAYMENT_status

DEFAULT 'Success',

CONSTRAINT PK_PAYMENT

PRIMARY KEY (payment_id),

CONSTRAINT UQ_PAYMENT_transaction_reference

UNIQUE (transaction_reference),

CONSTRAINT CK_PAYMENT_amount

CHECK (amount > 0),

CONSTRAINT CK_PAYMENT_method

CHECK

(

payment_method IN

('Card', 'EFT', 'Cash', 'Other')

),

CONSTRAINT CK_PAYMENT_status

CHECK

(

status IN

('Success', 'Failed', 'Refunded')

),

CONSTRAINT FK_PAYMENT_REGISTRATION

FOREIGN KEY (registration_id)

REFERENCES REGISTRATION(registration_id)

);

GO

-- 8. RACE_RESULT

CREATE TABLE RACE_RESULT

(

result_id INT IDENTITY(1,1) NOT NULL,

registration_id INT NOT NULL,

finish_time TIME(0) NULL,

overall_position INT NULL,

category_position INT NULL,

chip_time TIME(0) NULL,

status NVARCHAR(10) NOT NULL

CONSTRAINT DF_RACE_RESULT_status

DEFAULT 'Finished',

CONSTRAINT PK_RACE_RESULT

PRIMARY KEY (result_id),

CONSTRAINT UQ_RACE_RESULT_registration

UNIQUE (registration_id),

CONSTRAINT CK_RACE_RESULT_overall_position

CHECK

(

overall_position IS NULL

OR overall_position > 0

),

CONSTRAINT CK_RACE_RESULT_category_position

CHECK

(

category_position IS NULL

OR category_position > 0

),

CONSTRAINT CK_RACE_RESULT_status

CHECK

(

status IN

('Finished', 'DNF', 'DNS')

),

CONSTRAINT FK_RACE_RESULT_REGISTRATION

FOREIGN KEY (registration_id)

REFERENCES REGISTRATION(registration_id)

);

GO

-- 9. SPONSOR

CREATE TABLE SPONSOR

(

sponsor_id INT IDENTITY(1,1) NOT NULL,

sponsor_name NVARCHAR(100) NOT NULL,
contact_name NVARCHAR(100) NOT NULL,
email NVARCHAR(100) NOT NULL,
phone NVARCHAR(20) NOT NULL,
website NVARCHAR(200) NULL,
sponsorship_level NVARCHAR(20) NOT NULL,

CONSTRAINT PK_SPONSOR

PRIMARY KEY (sponsor_id),

CONSTRAINT UQ_SPONSOR_name

UNIQUE (sponsor_name),

CONSTRAINT UQ_SPONSOR_email

UNIQUE (email),

CONSTRAINT CK_SPONSOR_level

CHECK

(

sponsorship_level IN

('Gold', 'Silver', 'Bronze', 'Other')

)

);

GO

-- 10. RACE_SPONSOR

CREATE TABLE RACE_SPONSOR

(

 race_id INT NOT NULL,

 sponsor_id INT NOT NULL,

 CONSTRAINT PK_RACE_SPONSOR

 PRIMARY KEY (race_id, sponsor_id),

 CONSTRAINT FK_RACE_SPONSOR_RACE

 FOREIGN KEY (race_id)

 REFERENCES RACE(race_id),

 CONSTRAINT FK_RACE_SPONSOR_SPONSOR

 FOREIGN KEY (sponsor_id)

 REFERENCES SPONSOR(sponsor_id)

);

GO

-- INSERT ORGANIZERS

INSERT INTO ORGANIZER

(

organizer_name,

contact_name,

email,

phone,

website

)

VALUES

(

'RaceDay Events',

'Thabo Mokoena',

'thabo@raceday.co.za',

'0115550101',

'https://www.raceday.co.za'

),

(

'City Athletics Club',

'Lerato Dlamini',

```
'lerato@cityathletics.co.za',  
'0115550102',  
'https://www.cityathletics.co.za'  
);
```

GO

-- INSERT PARTICIPANTS / RUNNERS

INSERT INTO RUNNER

```
(  
    first_name,  
    last_name,  
    date_of_birth,  
    gender,  
    email,  
    phone,  
    address,  
    registration_date  
)
```

VALUES

```
(
```

'Mlungisi',
'Makhanya',
'2002-05-14',
'Male',
'mlungisi@example.com',
'0825551001',
'Johannesburg, Gauteng',
'2026-01-15'

),

(

'Amahle',
'Dlamini',
'2001-09-22',
'Female',
'amahle@example.com',
'0825551002',
'Soweto, Gauteng',
'2026-01-18'

),

(

'Kabelo',
'Molefe',
'1995-03-11',

```
'Male',  
'kabelo@example.com',  
'0825551003',  
'Pretoria, Gauteng',  
'2026-02-02'  
);
```

GO

-- INSERT LOCATIONS

INSERT INTO LOCATION

```
(  
    location_name,  
    address,  
    city,  
    province,  
    postal_code,  
    latitude,  
    longitude  
)
```

VALUES

(
 'Zoo Lake Sports Grounds',
 '57 Lower Park Drive',
 'Johannesburg',
 'Gauteng',
 '2193',
 -26.155300,
 28.026700

),

(
 'Loftus Versfeld Stadium',
 '440 Kirkness Street',
 'Pretoria',
 'Gauteng',
 '0002',
 -25.753300,
 28.226700

),

(
 'Emmarentia Botanical Garden',
 'Olifants Road',
 'Johannesburg',
 'Gauteng',

```
'2195',  
-26.155900,  
28.008000  
);
```

```
GO
```

```
-- INSERT 3 EVENTS / RACES
```

```
INSERT INTO RACE
```

```
(  
    race_name,  
    description,  
    race_date,  
    location_id,  
    race_type,  
    entry_fee,  
    capacity,  
    organizer_id,  
    status  
)
```

```
VALUES
```

(
 'Johannesburg City 10K',
 'Annual 10 kilometre road race through Johannesburg.',
 '2026-10-10',
 1,
 '10km',
 180.00,
 5000,
 1,
 'Scheduled'
),

(
 'Pretoria Spring Half Marathon',
 'A 21 kilometre road race celebrating the spring season.',
 '2026-10-25',
 2,
 '21km',
 250.00,
 4000,
 2,
 'Scheduled'
),

(

```
'Emmarentia Family 5K',  
'A family-friendly 5 kilometre community fun run.',  
'2026-11-07',  
3,  
'5km',  
100.00,  
2500,  
1,  
'Scheduled'  
);
```

GO

```
-- INSERT CATEGORIES FOR EACH EVENT
```

```
INSERT INTO RACE_CATEGORY
```

```
(  
    race_id,  
    category_name,  
    gender_eligibility,  
    age_from,  
    age_to
```

)

VALUES

-- Event 1

(

1,

'Open Male',

'Male',

18,

39

),

(

1,

'Open Female',

'Female',

18,

39

),

(

1,

'Masters Open',

'Open',

40,

99

),

-- Event 2

(

2,

'Senior Male',

'Male',

18,

39

),

(

2,

'Senior Female',

'Female',

18,

39

),

(

2,

'Masters Open',

'Open',

40,

99

),

-- Event 3

(

3,

'Junior Open',

'Open',

13,

17

),

(

3,

'Adult Open',

'Open',

18,

59

),

(

3,

'Family Open',

'Open',

13,

99

```
);
```

```
GO
```

```
-- INSERT SPONSORS
```

```
INSERT INTO SPONSOR
```

```
(
```

```
    sponsor_name,
```

```
    contact_name,
```

```
    email,
```

```
    phone,
```

```
    website,
```

```
    sponsorship_level
```

```
)
```

```
VALUES
```

```
(
```

```
    'MTN South Africa',
```

```
    'Ayanda Nkosi',
```

```
    'events@mtn.co.za',
```

```
    '0835552001',
```



```
'https://www.mtn.co.za',  
'Gold'  
,  
(  
  'Nedbank',  
  'Jason Smith',  
  'sports@nedbank.co.za',  
  '0835552002',  
  'https://www.nedbank.co.za',  
  'Gold'  
,  
(  
  'PUMA South Africa',  
  'Zanele Maseko',  
  'racing@puma.co.za',  
  '0835552003',  
  'https://za.puma.com',  
  'Silver'  
);
```

GO

-- =====

```
-- INSERT RACE-SPONSOR RELATIONSHIPS
```

```
-- =====
```

```
INSERT INTO RACE_SPONSOR
```

```
(
```

```
    race_id,
```

```
    sponsor_id
```

```
)
```

```
VALUES
```

```
(1, 1),
```

```
(1, 2),
```

```
(2, 2),
```

```
(2, 3),
```

```
(3, 1),
```

```
(3, 3);
```

```
GO
```

```
-- =====
```

```
-- INSERT SAMPLE ENROLMENTS / REGISTRATIONS
```

```
-- =====
```

```
INSERT INTO REGISTRATION
```

```
(  
    runner_id,  
    race_id,  
    registration_date,  
    status,  
    amount_paid,  
    payment_status,  
    bib_number  
)
```

VALUES

```
(  
    1,  
    1,  
    '2026-08-01',  
    'Confirmed',  
    180.00,  
    'Paid',  
    1001  
)
```

```
(  
    2,  
    1,  
    '2026-08-03',
```

'Confirmed',

180.00,

'Paid',

1002

),

(

3,

2,

'2026-08-05',

'Confirmed',

250.00,

'Paid',

2001

),

(

1,

2,

'2026-08-07',

'Confirmed',

250.00,

'Paid',

2002

),

```
(  
  2,  
  3,  
  '2026-08-10',  
  'Confirmed',  
  100.00,  
  'Paid',  
  3001  
);
```

GO

```
-- =====  
-- INSERT SAMPLE PAYMENTS  
-- =====
```

INSERT INTO PAYMENT

```
(  
  registration_id,  
  payment_date,  
  amount,  
  payment_method,  
  transaction_reference,
```

```
status
)
VALUES
(
    1,
    '2026-08-01T10:15:00',
    180.00,
    'Card',
    'RD-2026-0001',
    'Success'
),
(
    2,
    '2026-08-03T11:20:00',
    180.00,
    'EFT',
    'RD-2026-0002',
    'Success'
),
(
    3,
    '2026-08-05T09:45:00',
    250.00,
```

```
'Card',  
  
'RD-2026-0003',  
  
'Success'  
  
),  
  
(  
  
4,  
  
'2026-08-07T14:10:00',  
  
250.00,  
  
'EFT',  
  
'RD-2026-0004',  
  
'Success'  
  
),  
  
(  
  
5,  
  
'2026-08-10T16:30:00',  
  
100.00,  
  
'Card',  
  
'RD-2026-0005',  
  
'Success'  
  
);
```

GO

```
-- =====  
  
-- INSERT SAMPLE RESULTS  
  
-- =====
```

```
INSERT INTO RACE_RESULT
```

```
(  
    registration_id,  
    finish_time,  
    overall_position,  
    category_position,  
    chip_time,  
    status  
)
```

```
VALUES
```

```
(  
    1,  
    '00:52:34',  
    1,  
    1,  
    '00:52:10',  
    'Finished'  
),  
(
```



```
2,  
'00:58:21',  
8,  
4,  
'00:57:55',  
'Finished'  
);
```

GO

```
-- =====  
-- VERIFY THE DATABASE  
-- =====
```

```
SELECT * FROM RUNNER;  
  
SELECT * FROM ORGANIZER;  
  
SELECT * FROM LOCATION;  
  
SELECT * FROM RACE;  
  
SELECT * FROM RACE_CATEGORY;  
  
SELECT * FROM REGISTRATION;  
  
SELECT * FROM PAYMENT;  
  
SELECT * FROM RACE_RESULT;  
  
SELECT * FROM SPONSOR;
```

```
SELECT * FROM RACE_SPONSOR;
```

```
GO
```

